

MANDATORY-HANDS-ON-CTS-DN-5.0

Week 1

EcommercePlatform

FactoryMethodPatternExample

FinancialForecasting

JUnit Setup Examples

src

main

test

target

.gitignore

pom.xml

MockitoMockingExample

src

main

test/java/com/example

ExternalApi.java

MyService.java

MyServiceTest.java

target

pom.xml

PL SQL

SingletonPatternExample

.gitignore

OUTLINE

TIMELINE

JAVA PROJECTS

MAVEN

Week 1 > MockitoMockingExample > pom.xml

```
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
11 <dependencies>
22 <dependency>
26 | | | <scope>test</scope>
27 | | </dependency>
28 | </dependencies>
29 </dependencies>
30
31 <build>
32 <plugins>
33 <plugin>
34 | <groupId>org.apache.maven.plugins</groupId>
35 | <artifactId>maven-surefire-plugin</artifactId>
36 | <version>3.2.5</version>
37 </plugin>
38 </plugins>
39 </build>
40
41 </project>
```

PROBLEMS 10

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SQL HISTORY

TASK MONITOR

mathesh@Matheshs-MacBook-Air MockitoMockingExample % mvn clean test

tstrap classpath has been appended

WARNING: A Java agent has been loaded dynamically (/Users/mathesh/.m2/repository/net/bytebuddy/byte-buddy-agent/1.14.12/byte-buddy-agent-1.14.12.jar)

WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning

WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information

WARNING: Dynamic loading of agents will be disallowed by default in a future release

[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.355 s -- in com.example.MyServiceTest

[INFO] Results:

[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

[INFO] BUILD SUCCESS

[INFO] Total time: 4.342 s

[INFO] Finished at: 2026-06-25T11:32:15+05:30

mathesh@Matheshs-MacBook-Air MockitoMockingExample %

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

+ pom.xml

Describe what to build

+ | | Auto |

Local | Default Approvals

Ln 41, Col 11 Spaces: 2 UTF-8 LF () XML

MANDATORY-HANDS-ON-CTS-DN-5.0

Week 1

JUnit Setup Examples

src

test

MyServiceVerifyTest.java

Calculator.java

pom.xml

MyServiceVerifyTest.java 2, U

ExternalApi.java 2

My

CHAT

SESSIONS

Week 1 > MockitoMockingExample > src > test > java > com > example > J MyServiceVerifyTest.java > MyServiceVerifyTest

4 import org.junit.jupiter.api.Test;

5

6 public class MyServiceVerifyTest {

7

8 @Test

9 public void testVerifyInteraction() {

10

11 // Step 1: create mock

12 ExternalApi mockApi = mock(ExternalApi.class);

13

14 // Step 2: inject mock into service

15 MyService service = new MyService(mockApi);

16

17 // Step 3: call method

18 service.fetchData();

19

20 // Step 4: verify interaction

21 verify(mockApi).getData();

22 }

23 }

PROBLEMS 12

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SQL HISTORY

TASK MONITOR

mathesh@Matheshs-MacBook-Air Week 1 % mvn test

[INFO] --- surefire:3.5.4:test (default-test) @ junit_setup_examples ---

[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider

[INFO] -----

[INFO] T E S T S

[INFO] -----

[INFO] Running com.example.MainTest

[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.019 s -- in com.example.MainTest

[INFO] Results:

[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

[INFO] -----

[INFO] BUILD SUCCESS

[INFO] -----

[INFO] Total time: 0.667 s

[INFO] Finished at: 2026-06-25T11:49:54+05:30

[INFO] -----

mathesh@Matheshs-MacBook-Air Week 1 %

Tip: Try the Plan agent to research and plan before implementing changes.

+ J MyServiceVerifyTest.java

Describe what to build

+ | | Auto |

Local | Default Approvals

Ln 23, Col 2

Spaces: 4

UTF-8

LF

() Java